

Mohamed Nasir Mohamed

BSC APPLIED DATA SCIENCE COMMUNICATION (READING) · GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY

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Data Science undergraduate skilled in Python, SQL, R, and Power BI, with a strong foundation in machine learning, analytics, and AI systems. Co-author of a research paper accepted at ADScAI 2026 (University of Moratuwa). Currently building LaborEx, a WhatsApp-first labor platform for Sri Lanka's informal economy, and complementing academic work with 4+ years of hands-on retail experience. Seeking an internship in Data Science, AI Engineering, or ML Engineering.

EDUCATION

General Sir John Kotelawala Defence University <i>Bachelor's Degree – Applied Data Science Communication (Reading)</i>	Jan 2024 – Present
Al Azhar College <i>Local A-Levels – Engineering Technology, Science For Technology, ICT</i>	Jan 2020 – Dec 2022

ACADEMIC & PERSONAL PROJECTS

LaborEx – Blue-Collar Labor Exchange Platform <i>Full-Stack AI Platform</i>	2026 – Present <i>FastAPI, PostgreSQL, Python</i>
Built a WhatsApp-first labor exchange platform connecting informal daily-wage workers with SMEs across Sri Lanka Implemented PostGIS geo-matching, dual trust scoring system, and EPF/ETF compliance calculations Planned development of a multilingual NLP pipeline to handle Sinhala, Tamil, and English code switching for WhatsApp webhook integration. Technologies: FastAPI, PostgreSQL, PostGIS, WhatsApp Business API, Python	
USD/LKR Exchange Rate Prediction – ADScAI 2026 Conference Paper <i>Group Research Project (Accepted, University of Moratuwa)</i>	2026 <i>Python, Scikit-learn, XGBoost</i>
Co-authored a peer-reviewed research paper accepted at ADScAI 2026 conference (University of Moratuwa) on medium-term USD/LKR exchange rate forecasting Applied the CRISP-DM methodology end-to-end on macroeconomic data sourced from the Central Bank of Sri Lanka Compared classical ML models including Random Forest and XGBoost for medium-term (weeks/months) forecasting; performed feature engineering on lagged macroeconomic indicators Technologies: Python, Pandas, Scikit-learn, XGBoost, Random Forest	
Early Churn Detection & Customer Segmentation – Telecom ML Research <i>Group Research Project (IEEE-format Paper)</i>	2026 <i>Python, LightGBM, SHAP</i>
Co-authored a research paper proposing an early churn prediction framework (tenure ≤ 3 months) for the telecom sector using the IBM Telco dataset (7,043 customers) Designed methodology combining LightGBM (F1 = 0.973, ROC-AUC = 0.995), SMOTE class balancing, SHAP explainability, and KMeans customer segmentation Optimized decision threshold using the precision-recall curve and a business-cost model; identified three actionable churn segments to guide targeted retention strategies Technologies: Python, LightGBM, XGBoost, Scikit-learn, SHAP, SMOTE, KMeans	
PolicyAdapt AI – Policy Summarisation & Scenario Adaptation Platform <i>AI-Powered Policy Analysis System</i>	2026 <i>Python, Streamlit</i>
Built an end-to-end AI platform that ingests policy documents, extracts structured summaries using TextRank, and generates scenario adapted drafts using LLaMA 3.1 via GroqCloud Implemented multi-scenario drafting engine supporting audience, constraint, priority, and tone parameters with full draft versioning and history tracking Technologies: Python, Streamlit, LangChain, TextRank	
Agentic Research Paper Analyzer <i>AI Research Analysis System</i>	2026 <i>Python, Streamlit</i>
Developed a 4-step agentic RAG pipeline (Plan → Retrieve → Extract → Verify) to analyze academic papers with page level citations and anti hallucination design Built FAISS vector index with sentence transformer embeddings; implemented multi-paper literature review with cross-paper comparison Technologies: Python, Streamlit, FAISS, LangChain	

Advanced SQL & Cloud Databases: Government & SDG Data

Academic Project

Implemented CTEs, views, stored procedures, and cloud database integration

Technologies: T-SQL, SQL Server, Power BI

2025

SQL, Power BI

SQL & Power BI: Australian Government Data Analytics

SQL Project

Designed relational databases and complex T-SQL queries in SQL Server

Technologies: SQL, T-SQL, Power BI, DAX

2025

SQL, Power BI

EXPERIENCE

Sales Operations Associate (Part-Time / Flexible)

2025 – 2026

ACM Hardware (Pvt) Ltd., Mount Lavinia

Assisted in retail sales and customer engagement for a hardware products business serving diverse client segments.

Managed cashier operations, ensuring accurate cash handling and end-of-day billing reconciliation.

Developed strong technical communication skills explaining hardware products to non-technical buyers.

Built customer relationship workflows that later informed AI-driven automation project designs.

Sales Inventory Assistant (Part-Time)

2022 – 2025

Barakath Stores (Pvt) Ltd., Pilimathalawa

Supported sales operations, customer service, and inventory management across a high-volume retail store.

Tracked stock levels and analyzed sales trends to optimize product availability – directly inspired the Smart Retail Automation project.

Managed transactions and ensured accurate cash handling across daily operations.

Gained firsthand exposure to SME business management, supply chain decisions, and retail analytics.

TECHNICAL & INTEREST SKILLS

Programming: Python, R, SQL, Power BI

AI & ML: LangChain, RAG Systems, FAISS, Scikit-learn, LightGBM, XGBoost, SHAP, TextRank

Backend: FastAPI, PostgreSQL, PostGIS, WhatsApp Business API

Libraries: Pandas, NumPy, Matplotlib, Streamlit, Plotly, pypdf

Data Science: EDA, Data Cleaning, Feature Engineering, Model Evaluation, NLP, Time Series Forecasting

Database & Data Engineering: Relational Databases, ETL, Data Transformation

Tools: Git, GitHub, VS Code, Jupyter Notebook, Power BI, Microsoft Excel

Soft Skills: Leadership, Teamwork, Problem-Solving, Analytical Thinking, Adaptability

Interests: Agentic AI Systems, Data Engineering, Artificial Intelligence, Business Automation

CERTIFICATIONS & ACHIEVEMENTS

Research Paper Accepted – ADScAI 2026, University of Moratuwa (USD/LKR Exchange Rate Prediction)

Champions – Data Odyssey 2025

What is Data Science? – IBM / Coursera

Foundations of Data Science – Google

Foundations: Data, Data, Everywhere – Google

Develop Generative AI Applications: Get Started – IBM

Start Writing Prompts like a Pro – Udemy

Generative AI for Web Developers – LinkedIn Learning

LangChain for LLM Application Development – DeepLearning.AI

REFERENCES

Dr.Charith Silva

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